Yashikkumar Khunt

Software Developer · RAG, Context Engineering & Agentic LLM Systems

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PROFESSIONAL SUMMARY

Software developer and M.Sc. Computer Science candidate (Universität Paderborn) building RAG pipelines, typed context graphs and agentic multi-step LLM workflows. 2+ years as a full-stack Werkstudent shipping LLM agents, hybrid retrieval and data pipelines in Python and TypeScript; Master's thesis on conflict-aware context fusion for conversational decision support with an industry partner. Focused on context engineering, retrieval evaluation and turning AI into reliable, cost-aware product features.

EXPERIENCE

Master's Thesis – Context-Aware RAG & Agents · **Industry Partner (confidential)** *2026 – Present*

Python · RAG · Context Graph · Ollama · Meilisearch · Vector DB · Docker · RAGAS

- Designing a **typed multi-source context graph** on the partner's emergency-response data & AI platform, fusing user/session context, live operational data (weather, location, incident type, vehicle positions) and static corpora (SOPs, regulations) with provenance and reliability per fragment.
- Building a context-fusion RAG backend with **dynamic query routing** across vector, keyword and live-state stores, **priority** × **freshness** × **relevance ranking** via time-decay validity, and explicit conflict handling (prioritize, defer or abstain) injected into the prompt context.
- Building the evaluation harness: curated benchmark of 150–250 normal, stale-data and conflict queries; LLM-as-judge grounding, source attribution, abstention correctness, latency and token cost; ablations vs. flat-RAG baselines.

Werkstudent – Full-Stack Developer · **Safety Innovation Center GmbH** *May 2024 – Present*

Python · TypeScript · LlamaIndex · Ollama · Meilisearch · Node-RED · Angular · Docker

- Developed an LLM decision-support agent with LlamaIndex tool-calling over Ollama-hosted models and Meilisearch (BM25) retrieval, the base for the platform's AI features.
- Built and maintained ∼70 Node-RED data adapters ingesting and indexing security research funding portals across DE, FR, ES, NL and AT (incl. ANR ODS and NWOpen APIs).
- Standardized adapter documentation and status audits; built Angular smart city dashboards.

DApp Developer · **DOT3 Solutions** *Dec 2022 – Jul 2023*

Solidity · React.js · OpenZeppelin · ERC Standards · Web3.js

- Developed ERC-standard smart contracts and a React.js NFT airdrop platform; deployed an Ethereum e-voting system with Aadhar API authentication.

Earlier internships: Cloud Infrastructure Intern, Celebal Technologies (2022) – AWS/Azure IAM, firewall & VPN, CI/CD concepts · Flutter Developer Intern, Infikey Technologies (2021) – production Flutter app

SELECTED PROJECTS

EWC Rulebook RAG – Compliance-Grade Retrieval *2026*

Python · LangGraph · LangChain · ChromaDB · BM25 · React · Redux

- RAG over 25 rulebooks (734 pages) with article-aware chunking (2,382 chunks), hybrid dense + BM25 retrieval, and a multi-step LangGraph pipeline (route → retrieve → resolve → answer) with rule precedence, article-level citations and abstention when evidence is missing.

AutoCode – Autonomous Coding Agent *2026*

TypeScript · Fastify · React · PostgreSQL · Docker · Vercel AI SDK · Anthropic/OpenAI

- Self-hosted agentic developer tool driven from Discord that plans, edits code and ships fixes as PRs, using a multi-provider LLM tool-calling loop in hardened Docker sandboxes with guardrails, USD budgets and cost tracing.
- TypeScript monorepo (Fastify API, worker, React dashboard) on a PostgreSQL job queue, running 24/7 on Oracle Cloud.

SecAI – LLM-Assisted Secure Software Development *2024*

Java · SonarQube · CryptoAnalysis · LLMs – SAST platform explaining Java crypto misuses to developers

TECHNICAL SKILLS

RAG & Retrieval: Hybrid retrieval (dense + BM25, RRF), embeddings, chunking strategies, indexing pipelines, query routing, reranking, RAG evaluation (LLM-as-judge, RAGAS-style metrics, ablations)
Agents & Context: LangGraph, LangChain, LlamaIndex, Vercel AI SDK, tool-calling / ReAct agents, multi-step orchestration, context graphs, prompt & context engineering, guardrails, cost tracking
LLMs & Data Stores: Anthropic/OpenAI APIs, Ollama, HuggingFace; ChromaDB, Qdrant, Meilisearch, PostgreSQL
Languages & Frameworks: Python, TypeScript/JavaScript, Java, SQL, Swift, Solidity; Fastify, Node.js, React, Angular, Node-RED, GraphQL
DevOps & Tools: Docker, Kubernetes, GitHub Actions, Oracle Cloud, AWS/Azure IAM, pytest, Vitest, SonarQube, Linux
Languages (Human): English (C1), German (A2, currently learning), Gujarati (native), Hindi (fluent)

EDUCATION

M.Sc. Computer Science *2023 – Present*
Universität Paderborn, Germany *Thesis: Conflict-Aware Multi-Source Context Fusion for Conversational Decision Support*

B.Tech. Computer Engineering *2019 – 2023*
Charotar University of Science and Technology, India *CGPA: 9.52 / 10 (equivalent ECTS: 1.2)*